Soluble State Tracking in Materialized Fixed-Point GDN$^+$

Tarang Khanna
Independent Researcher
tarang.research@gmail.com

Abstract

Every finite soluble monoid admits a fixed-width, fixed-depth GDN$^+$ network computing nonempty prefix products in the materialized fixed-point model defined here, with constant gates $\alpha = \beta = 1$, zero initial memory, and either of two specified centered normalizations. Building on StateBench's aperiodic and rounded-parity constructions, we supply arbitrary-modulus counters and full-network soluble composition. Individually rounded materialized products implement bounded counters. Shifted copies yield constant-norm phase codes that survive normalization and residual addition; bias-free ReLU selectors return reusable labels. A rounded zero-state marker replaces a beginning-of-sequence token, and a parity-driven buffer supplies the exclusive prefix needed for wreath composition. Exhaustive certificates check complete modulo-three and $\mathbb{Z}_3 \wr U_2$ networks under both normalizations; the general result follows by algebraic induction. The construction depends on this evaluation program: the tested rounded rank-one evaluator fails with unchanged weights, without proving impossibility for rank-one models.

1 Introduction

State tracking asks a sequence model to apply successive updates without losing the information needed by later ones. Finite monoid word problems separate immutable assignment, mutable assignment, self-reference and general finite-state updates. In StateBench, Petty et al. [6] organize these as the strict hierarchy r-trivial $\subsetneq$ aperiodic $\subsetneq$ soluble $\subsetneq$ finite. Its Theorem 1 places fixed-precision GDN$^+$ above the aperiodic class, and Lemma 3 gives a rounded parity orbit. The exact placement relative to soluble tracking remains an open question in that paper.

The issue is not only whether a recurrent head can count. Its state must pass through legal projections, normalization, residuals and feedforward layers, then supply the timing and precision interfaces needed for composition. We construct these interfaces in a materialized rational-grid model.

Theorem 1 (Materialized fixed-point model)**.** *Fix either centered normalization and the scalar operations of §2, and permit constant gates $\alpha = \beta = 1$ as actual parameters. For every finite soluble monoid M, there exist fixed finite width, depth, head count and feedforward width, integers D, p, and rational parameters in the single datatype $\mathbb{D}_{p,D}$ such that the zero-initialized network returns the product of every nonempty word over M. Its canonical output at every position encodes the inclusive prefix product. All scalar parameters and pre-rounding intermediate results remain in range for arbitrary input length. There are no biases, positional inputs or beginning-of-sequence token. The two normalization conventions use different constructions.*

The theorem establishes containment of Sol in the defined model. The proof uses the usual Krohn–Rhodes induction; its architecture-specific content is in three primitives and their interfaces:

1. **Rescaled modular counters.** Separately rounding the m^2 matrix products implements a bounded counter for every fixed m. Rescaling the state and value makes the same integer transition available at compatible common denominators (§3).
2. **Normalization-exact carriers.** Shifted counter copies give a constant-norm phase code. Opposite shared coordinates cancel in the residual; a finite bias-free ReLU selector then returns a canonical label. A rounded zero-state marker initializes the shifts without a BOS token (§4; Figure 1; Table 1).
3. **Composable timing.** A parity-driven buffer exposes the previous lower prefix needed by a wreath product. Payload-preserving recoding and a single datatype close the induction (§5). Delayed composition itself is prior work [7, Lemmas 18, 20]; the construction here realizes it with the stated GDN$^+$ operations.

*Disclosure of AI assistance: large language models contributed to developing the mathematical constructions, implementing computational verification, checking mathematical arguments, and drafting this manuscript. The human author is responsible for the submitted claims and materials.

2 Model and arithmetic

We specify the following operational model, following the sublayer structure of Definitions 9, 10 and 13 in Petty et al. [6]. Tokens are embedded and normalized, $h_i^0 = \mathrm{LN}(E[w_i])$. Each layer computes

$$\hat{h} = \mathrm{LN}(h), \qquad g = h + \mathrm{LN}\big(O \cdot \mathrm{head}(\hat{h})\big), \qquad h' = g + W_2\ \mathrm{ReLU}(W_1\,\mathrm{LN}(g)),$$

Here head denotes the concatenated outputs of a fixed finite number of heads. Queries, keys and values are linear in $\hat{h}$, with no biases. The feedforward write is not normalized (Definition 10). A GDN head with one value column block maintains

$$A_t = I - \beta_t k_t k_t^\top, \qquad S_t = \alpha_t A_t S_{t-1} + \beta_t k_t v_t^\top, \qquad o_t = S_t^\top q_t,$$

with $S_0 = 0$. Throughout we permit constant $\alpha_t = \beta_t = 1$; with unit keys the unrounded A_t is a positive-semidefinite projector. This does not assert a spectral property of the rounded nonlinear transition map. The final output is $\arg\max h_n^d U$.

Arithmetic. Numbers live in $\mathbb{D}_{p,D} = \{j/D : \ j \in \mathbb{Z},\ |j| \le 2^p\} \cup \{\bot\}$. This rational grid makes the evaluation order discussed in StateBench Appendix A.4 explicit; identification with every inherited cast convention is not assumed. Before rounding, every scalar operation is checked against the physical range $[-2^p/D, 2^p/D]$; overflow produces absorbing $\bot$. Sums are exact unless they overflow. Host integers used to evaluate rational expressions are not themselves values in this datatype. A product $(a/D)(b/D)$ is rounded to $\mathrm{R}(ab/D)/D$ with $\mathrm{R}(x) = \lfloor x + \frac{1}{2} \rfloor$ (ties up). Iterated sums are evaluated left to right. Matrix products are evaluated entrywise with each scalar product rounded, and A_t is materialized before $A_t S_{t-1}$ is formed. Division and square root inside layer normalization are rounded to the grid in the same way; the construction arranges that every square-root argument is exactly 1, so only $\sqrt{1} = 1$ is used.

Normalization. For width n, let $\bar{x} = n^{-1} \sum_j x_j$. We construct different networks for

$$\mathrm{LN_E}(x) = \frac{x - \bar{x}\mathbf{1}}{\sqrt{\sum_j (x_j - \bar{x})^2}}, \qquad \mathrm{LN_V}(x) = \frac{x - \bar{x}\mathbf{1}}{\sqrt{n^{-1} \sum_j (x_j - \bar{x})^2}}.$$

There is no epsilon or affine term. A zero denominator produces $\bot$ and is never reached. Mean, square, root and division operations use the stated grid; all reached root arguments equal one. ReLU and argmax are exact on grid values.

Relation to StateBench. Definitions 9–10 fix the sublayer order used here. Definition 13 specifies closed gate ranges but also inherits the gate parameterization of Yang et al. [9]. StateBench itself uses endpoint gates in Lemmas 1b and 3, which supports an abstract endpoint interpretation; it does not by itself resolve the inherited activation/cast program. Its normalization inheritance includes a mean-zero Euclidean hash construction [4, Appendix C], while the complete fixed-grid centering/root/division path is not locally specified in Merrill et al. [3], Li et al. [2], Petty et al. [6]. Theorem 1 therefore states an unconditional result in the explicit operational model above. Identifying it with StateBench requires that model's gate and normalization interpretation to admit the same execution. This is a specification question, distinct from assuming or excluding finite-logit realizability in practical implementations.

3 Rounded cyclic counters

Lemma 1 (Counter)**.** *Let $D = Lm^2$ with $m \ge 2$, $L \ge 1$. Let k have m^2 coordinates equal to $1/m$ (so $\|k\| = 1$), $q = e_1$, $\alpha = \beta = 1$, and let one value column hold $v = a/L$ for an integer increment a. If every active coordinate of the state column equals $r/(Lm)$ for an integer r, and the scalar intermediates are in range, then after one step every active coordinate equals $r'/(Lm)$ with*

$$r' = r + a + m\ \mathrm{R}(-r/m) =: c_m(r) + a, \tag{1}$$

where $c_m(r) \in (-m/2,\, m/2]$ is the centered representative of $r \bmod m$.

Proof. In numerator units the state coordinate is mr. The materialized A has diagonal $1 - 1/m^2$ and off-diagonal $-1/m^2$. A diagonal product contributes $\mathrm{R}\left((1 - 1/m^2)\, mr\right) = mr + \mathrm{R}(-r/m)$ since mr is an integer; each of the $m^2 - 1$ off-diagonal products contributes $\mathrm{R}(-r/m)$. Summing the m^2 already-rounded terms gives $mr + m^2\,\mathrm{R}(-r/m)$. The write $k_i v = (1/m)(a/L)$ has numerator ma exactly. Dividing by m returns (1). □

Iterating (1) from r_0 gives $r_t \equiv r_0 + \sum_{i \le t} a_i \pmod m$ with $r_t \in (-m/2,\, m/2] + a_t$, a bounded orbit whose intermediates fit a sufficiently large fixed p (Appendix D). This uniform-row construction differs from StateBench's two-dimensional parity projector. Here $m^2 \mid D$, so $m = 2, D = 25$ is not an instance.

Remark (The rescaling must move the state). Raising D to Lm^2 while keeping the old state r/m and value a breaks the orbit. At $m = 3$, $D = 18$, $S_i = 1/3$, $a = 0$: the diagonal product has numerator $\mathrm{R}(16 \cdot 6/18) = 5$ and each of the eight off-diagonal products has numerator $\mathrm{R}(-12/18) = -1$, so the new coordinate is $(5 - 8)/18 = -1/6$ rather than $1/3$. The certificate checks this retained-update witness separately from the scaled identity; it does not claim a whole-network failure on a decoded word. The point matters because composing components forces a common denominator.

Constant and input channels in the same head. A value column receiving mb/L overwrites its state to b/L exactly: if the column previously held b'/L, the old grid numerator is $m^2 b'$, so its diagonal and off-diagonal contributions are $(m^2 - 1)b'$ and $-b'$, totaling zero. The write has numerator $m^2 b$, giving the new physical state b/L. We use this to carry the current increment a_t/L and the constant $1/L$ alongside the counters, so the output projection can form arbitrary fixed rational combinations of r_j, a_t and 1.

A first-position marker from the zero state. StateBench's Lemma 3 initializes the orbit with a beginning-of-sequence value. We avoid that token. A head with $k = \frac{1}{6}(-5, -3, -1, -1)$ (unit norm), $q = e_1$, $\alpha = \beta = 1$ and two value columns $2/D$, $1/D$ has active transition $A = I - kk^\top = \frac{1}{36}\left(\begin{smallmatrix} 11 & -15 & -5 & -5 \\ -15 & 27 & -3 & -3 \\ -5 & -3 & 35 & -1 \\ -5 & -3 & -1 & 35 \end{smallmatrix}\right)$. Writing T for its rounded numerator action, direct evaluation gives, with $u = (-2, -1, 0, 0)$, $v = (-3, -1, 0, 0)$, $w = (-1, 0, 0, 0)$: $\mathrm{R}(2k) = u$, $\mathrm{R}(k) = w$, $T(u) = T(v) = w$, $T(w) = 0$. The two columns therefore follow $0 \to u \to v \to v \to \cdots$ and $0 \to w \to w \to \cdots$. Reading the first coordinates σ_t, γ_t,

$$B_t := D\sigma_t - 3D\gamma_t = \mathbf{1}[t = 1], \qquad C_t := -D\gamma_t = 1 \quad (t \ge 1).$$

Both are linear in the head output. The orbit uses the tie rule at $-1/2$; under ties-away-from-zero the displayed stationary orbit is not preserved.

4 A layer-norm-exact interface

A scalar counter readout does not by itself supply the reusable constant-norm interface required by the following construction. The fix is to run m counters whose joint state has constant norm.

Phase codes. Let $u_s = \big(c_m(s + j)\big)_{j=0}^{m-1}$ for $s \in \mathbb{Z}_m$ and $H = \sum_j c_m(j)^2$. The u_s are distinct cyclic shifts of one integer vector, with $\|u_s\|^2 = H$. For $s \ne t$, Cauchy–Schwarz gives $u_s^\top u_t \le H$, and equality would force $u_s = u_t$. Thus the integer $u_s^\top u_t$ is strictly below H, hence at most $H - 1$; equivalently $H - u_s^\top u_t = \|u_s - u_t\|^2/2 \ge 1$. For $m = 3$: $u_0 = (0, 1, -1)$, $u_1 = (1, -1, 0)$, $u_2 = (-1, 0, 1)$, $H = 2$.

First use a marker layer and a finite recoder to retain the current label and B_t. Run m counter columns $j = 0, \ldots, m - 1$ with values $v_j = (a_t + c_m(j)B_t)/L$. The marker fires once, at $t = 1$, so $r_{j,1} = a_1 + c_m(j)$, and thereafter each column follows (1). By induction,

$$\big(r_{j,t} - a_t\big)_{j=0}^{m-1} = u_{s_{t-1}}, \qquad s_t = \sum_{i \le t} a_i \bmod m,\ s_0 = 0. \tag{2}$$

The head output $S_t^\top q$ thus exposes $r_{j,t}/(mL)$, a_t/L and $1/L$; the output projection O forms $(r_{j,t} - a_t)/(H+1)$ with integer coefficients $mL/(H + 1)$ and $-L/(H + 1)$ once L is divisible by $H + 1$.

Two complete layers, width 28; indices below are zero-based

$$x_{a_t} \longrightarrow \underbrace{\text{LN} \;\to\; \text{marker head} \;\to\; O \;\to\; \text{LN} \;\to\; +x_{a_t} \;\to\; \text{LN} \;\to\; \text{FF} + g}_{\text{layer 1: } S_0=0,\ B_t=\mathbf{1}[t=1]} \longrightarrow x_{(a_t,B_t)}$$

$$\underbrace{\text{LN} \;\to\; \text{shifted counters} \;\to\; O \;\to\; \text{LN} \;\to\; +x_{(a_t,B_t)} \;\to\; \text{LN} \;\to\; \text{FF} + g}_{\text{layer 2: exclusive phase } c=u_{s_{t-1}}} \longrightarrow x_{s_t} \xrightarrow{U} s_t$$

Coordinates	Count	Layer-2 output write y
0–11: six token-pair slots	12	zero
12–17: three phase pairs	6	$(c_0, -c_0, c_1, -c_1, c_2, -c_2)/4$
18–21: two padding pairs	4	$(1, -1, 1, -1)/4$
22–25: unused	4	zero
26–27: shared pair	2	$(-1/2, +1/2)$
Total	28	$\|y\|^2 = 1,\ \sum_i y_i = 0$

Input x has one active token pair and shared $(+1/2, -1/2)$.
In $g = x + y$, the shared pair cancels; token, phase and padding squared norms are 1/2, 1/4, 1/4. The FF selector writes $x_{s_t} - g$.

Figure 1: Coordinate-exact Euclidean implementation at $D = 1008$, $L = 112$, $p = 21$. Each FF is $W_2\,\mathrm{ReLU}(W_1\,\mathrm{LN}(g))$ followed by its residual addition; its write is not normalized. In layer 1, coordinates 12–15 instead hold the two marker-label pairs, and 16–25 are zero. The FF returns a canonical label, clearing that workspace before layer 2 reuses it. Each layer has one head. The source checker verifies these slots and the worked trace in Table 1 through the actual matrix implementation.

Canonical carrier. For a finite label alphabet Z, encode label z as

$$x_z = \tfrac{1}{2}\,(e_z,\ -e_z;\ 1,\ -1),$$

a centered unit vector: $\mathrm{LN}_{\mathrm{E}}(x_z) = x_z$. Figure 1 and its certificate interleave each positive coordinate with its negative partner, a fixed permutation of the displayed block notation. Its last two coordinates are a shared constant pair; $2(x_z)_{\mathrm{shared},+} = 1$ supplies every constant the projections need, and $2(x_z)_{\mathrm{token},z'} = \mathbf{1}[z = z']$ supplies every label indicator. Let the sublayer write be

$$y_s = \Big(\tfrac{u_s}{H+1},\ -\tfrac{u_s}{H+1};\ \tfrac{H-1}{2(H+1)},\ -\tfrac{H-1}{2(H+1)};\ -\tfrac{1}{2},\ \tfrac{1}{2}\Big),$$

in disjoint coordinate regions (phase pair, padding pair, shared pair). Its private part has squared norm $2H/(H+1)^2 + (H-1)^2/(2(H+1)^2) = \frac{1}{2}$, so $\|y_s\| = 1$, y_s is centered, and $x_z^\top y_s = -\frac{1}{2}$ through the shared pair. Hence

$$g = x_z + y_s \quad \text{satisfies} \quad \textstyle\sum_j g_j = 0, \quad \|g\|^2 = 1 + 1 - 1 = 1.$$

All three normalizations in the layer, $\mathrm{LN}(h)$, $\mathrm{LN}(O\,o)$ and $\mathrm{LN}(g)$, receive centered unit vectors and return them unchanged; in the fixed-point implementation every mean is zero and every squared Euclidean norm is exactly 1 (the population variance is $1/n$), and the only root taken is $\sqrt{1}$. Normalization is executed, not bypassed.

Decoder. The hidden unit

$$J_{s,z}(g) = \mathrm{ReLU}\left((H+1)\,u_s^\top g_{\mathrm{phase}} + 2g_{\mathrm{token},z} - 2H\textstyle\sum_{z'} g_{\mathrm{token},z'}\right)$$

has pre-activation $u_s^\top u_{s_{t-1}} - H + \mathbf{1}[z = z_t]$, which equals 1 when $(s, z) = (s_{t-1}, z_t)$ and is ≤ 0 otherwise. The $m|Z|$ units are therefore exactly one-hot without a bias. Setting the W_2 column for (s, z) to $x_{F(s,z)} - g(s, z)$, for any finite recoding F, gives $h' = g + W_2 J(g) = x_{F(s_{t-1},z_t)}$. Taking $F(s, z) = (z,\ s,\ s + a(z) \bmod m)$ preserves the input label and both the exclusive and inclusive prefix, which is what composition needs. The unembedding $U_r(h) = 2h_{\mathrm{token},r}$ yields one logit 1 and the rest 0.

t	a_t	B_t	(r_0, r_1, r_2)	$r_t - a_t\mathbf{1} = u_{s_{t-1}}$	s_t
1	1	1	$(1, 2, 0)$	$(0, 1, -1)$	1
2	1	0	$(2, 0, 1)$	$(1, -1, 0)$	2
3	2	0	$(1, 2, 3)$	$(-1, 0, 1)$	1

Table 1: Complete-network trace on $(1, 1, 2)$. The first-position bit initializes the three offsets once. The phase is the exclusive prefix code; the decoded residue s_t is inclusive. The same scalar recurrence independently gives $r_{j,1} = a_1 + c_3(j)$ and $r_{j,t} = a_t + c_3(r_{j,t-1})$ thereafter.

Centered unit variance. Under $\mathrm{LN_V}$ the fixed points are centered vectors of squared norm n. Uniformly scaling the Euclidean construction by $\sqrt{n}$ can leave no coordinate from which the tiny constants $1/D, 2/D$ of the marker head are legally generated. Instead take $n = 4\rho^2$, token pair ± 1, fixed private pairs whose squared amplitudes sum to $\rho^2 - 1$ and include a permanent $+1$ coordinate (a four-square decomposition supplies them), and shared pair $\pm\rho$; scale the Euclidean private write by 2ρ and reverse the shared pair. Use disjoint input-private and output-private coordinate regions. Then $\|x\|^2 = \|y\|^2 = \|g\|^2 = n$, the selector becomes $\mathrm{ReLU}\left(\frac{H+1}{2\rho} u_s^\top g_{\mathrm{phase}} + g_{\mathrm{token},z} - H g_{\mathrm{const}}\right)$ with the same integer margin, and the $+1$ coordinate generates all small constants. The concrete $m = 3$ network uses $n = 36$, amplitudes $(2, 1, 1, 1, 1)$, shared ± 3.

Concrete $\mathbb{Z}_3$ network. Two layers (marker, then phase), one head each, width 28, feedforward width 18, $D = 1008$, $L = 112$, $p = 21$ for $\mathrm{LN_E}$; width 36, $p = 24$ for $\mathrm{LN_V}$. The variance construction needs the larger numerator range: at $p = 21$, a parameter grid numerator 3,048,192 exceeds 2^{21}.

5 Composition and the exclusive prefix

Flip-flop. For $U_2 = \{1, a, b\}$ use two state rows, $q = e_1$, key e_0 on the identity and e_1 on a, b, values $0, e_a, e_b$, $\alpha = \beta = 1$. Row 0 stays zero; the identity leaves row 1 untouched and a reset overwrites it. With the constant channel, O forms the one-hot $(1 - S_a - S_b,\ S_a,\ S_b)$ including the initial identity. A separate overwrite head supplies that constant. The recurrent update uses only $\{0, \pm 1\}$; the carrier uses the rational grid specified above. (StateBench's Lemma 1b uses $\beta = 0$ on the identity; we use key selection instead so that every head in the network has $\alpha = \beta = 1$.)

Exclusive-prefix interface. For $W = M \wr N$ with input letters (f_t, n_t) and prefix (F_t, N_t), Definition 4 of Petty et al. [6] gives

$$N_t = N_{t-1} n_t, \qquad F_t(x) = F_{t-1}(x)\, f_t(x N_{t-1}).$$

The upper increment depends on the lower prefix *excluding* the current letter. The proof of Lemma 2 proposes a feedforward lookup from represented partial products. If these carry only the inclusive N_t, the following obstruction applies to that connector.

Proposition 1. *No function of (f_t, n_t, N_t) computes $f_t(xN_{t-1})$ in general. Over $\mathbb{Z}_3 \wr U_2$, let $f(1) = f(a) = 0$, $f(b) = 1$ and let 0 denote the constant-zero function. The words $(0, a)(f, a)$ and $(0, b)(f, a)$ have the same second letter and the same inclusive prefix $N_2 = a$, yet the upper increments at position 2 are $f(a) = 0$ and $f(b) = 1$.*

The displayed pair contradicts the existence of that lookup at $x = 1_N$. It establishes a missing interface obligation for an inclusive-only proof; it does not show the lemma's depth bound is unattainable by some other construction, and we do not label the full lemma an erratum.

Repair: a legal one-step delay. Run a $\mathbb{Z}_2$ phase component on constant increment 1 to obtain $p_t = t \bmod 2$ as a label. A buffer head with $k_t = e_{p_t}$, $q_t = e_{1-p_t}$, $v_t = e_{N_t}$, $\alpha = \beta = 1$ overwrites row p_t and reads row $1 - p_t$, so $S_t^\top q_t = 0$ at $t = 1$ and $e_{N_{t-1}}$ for $t \geq 2$; using a separate overwrite companion for the current B_t and constant, adding $B_t e_{1_N}$ from the marker gives $e_{N_{t-1}}$ for all $t \geq 1$. All multipliers are 0 or 1. The decoder of §4 then emits a canonical label carrying (f_t, n_t, N_t, N_{t-1}).

Closure. Given complete components for M and N that preserve any finite payload, the W component computes N_t, delays it, and for each fixed $x \in N$ runs a copy of the M component on the increment $f_t(xN_{t-1})$, which is a finite function of the current canonical label and is realized by the same projection-plus-decoder pattern. Copies run sequentially in depth, so no two normalized carriers are ever added. The depth is finite and independent of input length; no sharp depth bound is asserted. For a divisor A of W (a quotient of a submonoid $T \leq W$), choose any section $\sigma : A \to T$ of the quotient map π; run the W program on $\sigma(a_t)$ and decode π; multiplicativity of σ is not needed since $\pi(\prod \sigma(a_i)) = \prod a_i$. With the soluble Krohn–Rhodes decomposition into U_2 and cyclic-prime factors [6, Prop. 1, Def. 8], this proves Theorem 1.

One datatype. Fix the program, alphabets, and a common width and head count first (padding with zero heads). Choose D divisible by 36, by every factor's m^2, and by every carrier, projection and normalization denominator, and let $L_m = D/m^2$ with $(H_m + 1) \mid L_m$. The fixed rational ordinary products are exact after clearing their individual denominators; $1/D$ recurrent products require the separate primitive audit of Appendix D. In particular, the only intentionally rounded products are those of Lemma 1 and the marker, whose rounded maps are independent of D by construction. All state sets are finite and all scalar computations have fixed length, so a finite p bounds every parameter and every pre-rounding partial result for all input lengths.

6 Exact verification

For each concrete instance below, a certificate builds $E, Q, K, V, O, W_1, W_2, U$, executes the state matrices with the arithmetic of §2 (including the normalization primitives, which are computed, not replaced by the identity), rejects any pre-rounding grid numerator outside $[-2^p, 2^p]$, and exhausts the reachable transition system from $S_0 = 0$. Structural zero products are omitted while preserving the order of nonzero products. Because the state space is finite and closed, closure certifies all nonempty input lengths for these concrete networks. The certificate implementations were originally written from the mathematical specification without reading the original construction implementation; this is independence of implementation, not of invention.

Network	Convention	Width / D / p	States / transitions
$\mathbb{Z}_3$ (2 layers)	$\mathrm{LN_E}$	28 / 1008 / 21	13 / 39
$\mathbb{Z}_3$ (2 layers)	$\mathrm{LN_V}$	36 / 1008 / 24	13 / 39
$\mathbb{Z}_3 \wr U_2$ (7 layers, 81 letters)	$\mathrm{LN_E}$	672 / 96768 / 64	5,104 / 413,424
$\mathbb{Z}_3 \wr U_2$ (7 layers, 81 letters)	$\mathrm{LN_V}$	784 / 7056 / 64	5,104 / 413,424

For each $\mathbb{Z}_3$ arm: 390 normalization-routine calls (including verifier-only canonical fixed-point checks), all with exact zero mean and exact unit norm (resp. variance); no overflow; all 9,840 nonempty words of length ≤ 8 and 200 random words of length 500–5000 decoded correctly at every prefix. Short replay checks cover 73,812 positions and long replay checks 538,724 positions per arm, using seed 20261108. They replay transitions already evaluated through the full fixed-point network; they are not extra matrix evaluations. Wreath verification also memoizes complete local layer transitions; all 413,424 global edges are checked against a separately computed reference product. The maximum pre-rounding grid numerators are 1,524,096 and 9,144,576 for the two $\mathbb{Z}_3$ arms, and 14,046,068,736 and 2,091,059,712 for the respective wreath arms; all are below their displayed 2^p bounds. The $\mathbb{Z}_3 \wr U_2$ certificates include both reset witnesses of §5 and 22,977 normalization calls per arm, all exact. The general-modulus algebra of §4 (phase identities, norm identities, selector margins) was checked for $m = 2, \ldots, 13$ (818 phase pairs); the proof for general m is the induction above, not the enumeration.

Which assumptions carry the weight. Re-running the $\mathbb{Z}_3$ certificates with ties-to-even, or with a floored square root, passes (every root argument is 1). Fused-exact evaluation instead computes the rank-one affine recurrence with exact intermediates and one cast per state output. It first breaks the marker canonical invariant on $[0, 0]$, before the final cyclic readout. Ties-away-from-zero breaks the marker orbit. These two negative witnesses concern intermediate invariants, not demonstrated incorrect final modulo answers. The passing ties-even arm shows that the baseline tie rule is not necessary for this particular network.

Complete-output rank-one sensitivity. A separate alternative evaluates the rank-one update with individually rounded products. Write K_i, S_{ij}, V_j for integer grid numerators and compute

$$d_j = \sum_{i=0}^{q-1} \mathrm{R}(K_i S_{ij}/D),$$
$$S'_{ij} = \big(S_{ij} - \mathrm{R}(K_i d_j/D)\big) + \mathrm{R}(K_i V_j/D).$$

Here q is the number of active key rows. Sums are checked left to right; subtraction precedes write addition. The matrix $I - kk^\top$ is not materialized. All other operations, normalization checks and overflow checks stay unchanged. This is not the fused-exact evaluator above. Complete runs from zero memory under both normalization constructions give the same table:

Input word	Materialized outputs	Rounded rank-one outputs
$[1, 1]$	$[1, 2]$	$[1, 1]$
$[1, 2]$	$[1, 0]$	$[1, 2]$
$[1, 0]$	$[1, 1]$	$[1, 0]$

The certificate executes twelve complete network runs, with no overflow. The mechanism is explicit: for a uniform counter column of numerator mr, each rounded product in $k^\top S$ has numerator r, giving $d = m^2 r$; the correction has numerator mr and cancels the retained state exactly. For the marker, $\sum_i \mathrm{R}(k_i u_i) = 3$ and $\mathrm{R}(3k) = u$, while $\sum_i \mathrm{R}(k_i w_i) = 1$ and $\mathrm{R}(k) = w$. Thus the alternative repeats its first-position column states, reports $B_t = 1$, and reinitializes the phase at each step. This explains the current-increment outputs in the table. These results concern the specified evaluators, not production kernels.

7 Discussion

Relation to prior results. Petty et al. [6, Theorem 1; Lemmas 1b–3] supply aperiodic and rounded-parity baselines for GDN$^+$; their Theorem 2 gives a soluble lower bound for log-precision averaging hard attention. We construct arbitrary-modulus counters, normalized payload-preserving interfaces and complete soluble closure in the explicit materialized model. Our exclusive-prefix obstruction concerns a particular inclusive-only connector, not the attainability of the cited composition lemma's depth bound. Grazzi et al. [1, Theorem 1; Appendix A.4] prove a nonnegative-spectrum parity obstruction under a different exact-intermediate/casting convention. Separate scalar casts change the transition, so there is no contradiction under shared assumptions. Siems et al. [8, Theorems 1–2; Appendix B.1] obtain broad group and all-regular expressivity with DeltaProduct, including one update per head at sufficient depth, with coefficients up to 2, abstract parameter maps and decoder, and task-dependent initialization or a BOS token. These results do not supply the present zero-state, constant-gate, individually rounded interfaces. Nowak et al. [5, Appendix J] provide the arithmetic-sensitive algebraic framework and diagonal-recurrence cases. Unsigned-integer wrap-around differs from the present signed fixed-grid rounding. Sarrof et al. [7, Lemmas 18, 20] already implement previous-token access and delayed composition for SSMs. Neither rounding-induced periodicity, Krohn–Rhodes decomposition nor delayed composition is claimed as new here.

NFSM [10, Definitions 2.1–2.2] requires correct decoding along every trajectory with arbitrary per-step perturbations bounded by a positive radius. Our theorem concerns one deterministic rounding program, not that robustness guarantee. Its rounded state updates are nonlinear maps, so the affine robustness obstruction does not contradict this result. Complex KDA [11, Theorems 3, 5; Appendix B.4] uses signed channel gates and extended delta coefficients; its constructive arithmetic uses exact datatypes, potentially containing algebraic irrationals. Neither those results nor ours transfer automatically to floating-point execution.

Construction cost. The cyclic head represents $m^2(m+2) = \Theta(m^3)$ active recurrent entries. Naive dense evaluation of its retained product uses $m^4(m+2) = \Theta(m^5)$ scalar multiplications per token: $m^2(m+2)$ dot products of length m^2. These are active-block counts, not minimal storage, bit complexity or full padded-network cost. Composition adds $|N|$ sequential upper blocks, a clock, delay and finite controllers to the lower block; no optimal-depth or polynomial-size bound is asserted.

Limitations. Exact StateBench identification remains open. This is an existence bound without an efficient-size guarantee, upper bound, empty-word result, or consequence for nonzero epsilon, practical kernels or trained models. Finite certificates validate concrete networks; the general theorem is algebraic. The inclusive-prefix witness does not refute Lemma 2's statement or depth bound.

Reproducibility. Repository: `https://github.com/TarangKhanna/gdn-plus-soluble`. The verification implementation is fixed at commit `b4c080f9115598af59d0e0543ee7fbb704283e08`. The accompanying LaTeX source package contains its exact export at `anc/gdn_plus_verification_code.zip`; the repository's `paper/` directory contains this revision's PDF and source. The original Zenodo preprint (DOI: 10.5281/zenodo.24057440) and the immutable original code baseline retain the earlier provenance. The current implementation exposes the constant-gate wreath construction directly and adds the complete-output rank-one certificate; original expected results remain unchanged. From a clean extraction of the code archive, run `python3 -B certs/run_all.py`, without `-O`. The full suite passed on macOS with Python 3.14.4; no new Linux or Python 3.10 verification is claimed. Code is standard-library only. The general theorem rests on the proof appendices, not the finite certificates. AI assistance in construction, verification, review and drafting is disclosed on the first page; the author retains responsibility for the claims and final revision.

References

[1] R. Grazzi, J. Siems, A. Zela, J. Franke, F. Hutter, M. Pontil. Unlocking state-tracking in linear RNNs through negative eigenvalues. ICLR 2025; arXiv:2411.12537v4.

[2] Y. Li, A. Svete, A. Sabharwal, W. Merrill. Efficiently representing algorithms with chain-of-thought transformers. arXiv:2606.19697, 2026.

[3] W. Merrill, H. Jiang, Y. Li, A. Lin, A. Sabharwal. Why are linear RNNs more parallelizable? arXiv:2603.03612, 2026.

[4] W. Merrill, A. Sabharwal. The expressive power of transformers with chain of thought. arXiv:2310.07923v5.

[5] F. Nowak, R. Cotterell, R. Boumasmoud. An algebraic view of the expressivity of recurrent language models. ICML 2026; arXiv:2606.01765v3.

[6] J. Petty, W. Merrill, A. Sabharwal. A fine-grained hierarchy and benchmark for state tracking in language models. arXiv:2611.03094v1, 2026.

[7] Y. Sarrof, Y. Veitsman, M. Hahn. The expressive capacity of state space models: a formal language perspective. NeurIPS 2024.

[8] J. Siems, T. Carstensen, A. Zela, F. Hutter, M. Pontil, R. Grazzi. DeltaProduct: improving state-tracking in linear RNNs via Householder products. NeurIPS 2025; arXiv:2502.10297v7.

[9] S. Yang, J. Kautz, A. Hatamizadeh. Gated delta networks: improving Mamba2 with delta rule. arXiv:2412.06464, 2024.

[10] J. Brandoit, A. Fyon, T. Braipson, T. Clara, F. De Geeter, P. Sacré, D. Ernst, G. Drion. Length-independent state tracking under a parallel scan. arXiv:2609.32447v1, 2026.

[11] J. Siems, R. Grazzi, K. Pöppel, J. Singh, A. Zela, T. Carstensen, J. Jitsev, F. Hutter, V. Cevher, A. Orvieto, A. Klein. Complex KDA: understanding and enhancing the expressivity of Kimi Delta Attention. arXiv:2609.24797v1, 2026.

A Primitive constructions and exact arithmetic

This appendix gives the local implementation steps used by the induction. A label always denotes an entire finite tuple, encoded with one active token pair. Enlarging the tuple alphabet enlarges width; it does not concatenate several independently normalized labels.

A.1 Projection and finite recoding

For Euclidean canonical input x_z, a linear map with column $2b_z$ at the positive token coordinate returns any prescribed rational vector b_z on label z. The shared positive coordinate supplies a constant by weight twice that constant. After choosing the grid as below, each individual product is representable; no affine bias is used.

A finite function $F : Z \to B$ can be computed as a complete layer. Use the unit-key overwrite head $k = q = e_0$, $v = (e_{F(z)}, 1)$, so the readout is the current one-hot and constant, regardless of previous memory. Let its output projection emit a private pair $(e_b/2, -e_b/2)$ and shared pair $(-1/2, 1/2)$. The residual contains the input-token pair and output-private pair with the shared pair canceled. For each (z, b), the bias-free selector

$$J_{z,b}(g) = \mathrm{ReLU}\left(2g_{\mathrm{token},z} + 2g_{\mathrm{output},b} - 2\sum_{z'} g_{\mathrm{token},z'}\right)$$

is exactly one for the actual pair and zero otherwise. Its W_2 column is $x_{G(z,b)} - g(z, b)$, where G can retain any original payload together with $F(z)$. This is a legal finite controller, not an external runtime table.

The marker uses the same output-pair geometry with $b = B_t$. Its two value columns are generated from the shared $+1/2$ by weights $4/D$ and $2/D$. For unit-variance inputs the permanent $+1$ generates them directly. The marker supplies B_t at every position; a recoder can retain it with the entire incoming label. Subsequent layers can recompute or preserve it.

A.2 Cyclic initialization and boundedness

Let the finite label determine $a_t \in \{0, \ldots, m-1\}$. Define $c_m(r) = r + mR(-r/m)$. This is periodic modulo m, congruent to r, and takes values in $(-m/2, m/2]$. For the shifted column initialized with the marker,

$$r_{j,1} = a_1 + c_m(j), \qquad r_{j,t} = a_t + c_m(r_{j,t-1}) \quad (t \geq 2).$$

Assume $r_{j,t-1} = a_{t-1} + c_m(s_{t-2} + j)$. Periodicity and congruence give $c_m(r_{j,t-1}) = c_m(s_{t-1} + j)$, proving the claimed exclusive phase. This also proves that the raw integer states range over a fixed finite set, independent of length. At the first step the same identity holds with $s_0 = 0$. The companion columns with values ma_t/L and m/L supply the current input and constant by exact overwrite, as calculated in the main text.

For completeness, the marker's row action is $T(z)_i = \sum_j R(A_{ij} z_j)$ in integer numerator units. Direct substitution in its displayed 4×4 matrix gives

$$R(2k) = u, \quad R(k) = w, \quad T(u) = w, \quad T(v) = w, \quad T(w) = 0, \qquad v = u + w.$$

Thus its memory invariant is exactly the two finite column orbits given in the main text; the initialization correction is available at all lengths.

A.3 The concrete modulo-three Euclidean carrier

The generic phase carrier uses denominator $H + 1$ and is convenient for the proof. The width-28 executable certificate uses a different, equally exact private geometry: $c = u_s \in \{(0, 1, -1), (1, -1, 0), (-1, 0, 1)\}$ and

$$y = \big(c/4, -c/4; 1/4, -1/4; 1/4, -1/4; -1/2, 1/2\big).$$

The first two padding pairs have total squared norm $1/4$, the phase pairs have squared norm $1/4$, and the shared pair has squared norm $1/2$. After adding the input token pair, the shared pair cancels and the residual again has squared norm one. The head reads $r_j/(3L), a/L, 1/L$; output weights $3L/4, -L/4$ form $(r_j - a)/4$. Padding and shared coordinates use weights $\pm L/4, \pm L/2$. For label $\ell = (a, B)$ and padding coordinate $1/4$,

$$J_{s,\ell}(g) = \mathrm{ReLU}\big(4u_s^\top g_{\mathrm{phase}} + 2g_{\mathrm{token},\ell} - 8g_{\mathrm{pad}}\big)$$

is exactly the joint indicator. Its W_2 column is $x_{(s+a) \bmod 3} - g_{s,\ell}$, where $g_{s,\ell}$ is the reached residual for that joint label. The residual addition therefore returns $g_{s,\ell} + (x_{(s+a) \bmod 3} - g_{s,\ell}) = x_{(s+a) \bmod 3}$, clearing the phase and padding coordinates. All 18 hidden units are explicit and bias-free. The last unembedding is $U_r(h) = 2h_{\mathrm{token},r}$, giving a unique unit logit. The variance certificate has its own carrier and selector constants; it is not obtained merely by changing the normalizer in this Euclidean network.

B Variance-one geometry at arbitrary width

Fix the maximum number K of joint labels and the maximum number J of private-output coordinate slots in the finite program. Choose an integer $r \geq 2$ so that $n = 4r^2$ accommodates all token, fixed-private, output-private and shared slots, with these regions disjoint. By the four-square theorem, write $r^2 - 2 = b_1^2 + \cdots + b_4^2$. The canonical input has one active token pair ± 1 among K pairs, fixed-private pairs of amplitudes $1, b_1, \ldots, b_4$, and shared pair $\pm r$. The first fixed pair includes a permanent $+1$. Its input-private squared norm is

$$2 + 2(1 + b_1^2 + \cdots + b_4^2) = 2r^2,$$

and its shared squared norm is $2r^2$, giving $\|x_z\|^2 = n$ and zero mean. Only finitely many slots are needed, while $4r^2$ can be arbitrarily large.

Scale any Euclidean private write (squared norm $1/2$) by $2r$, and use the opposite shared pair $\mp r$. Then $\|y\|^2 = n$ and $x_z^\top y = -2r^2$. The residual therefore has squared norm n, population variance one and zero mean. Its permanent $+1$ input coordinate survives. The normalization root argument is one, even though the squared Euclidean norm is n.

For a paired one-hot private output of amplitude r, the selector is

$$\mathrm{ReLU}(g_{\mathrm{token},z} + g_{\mathrm{output},b}/r - g_{\mathrm{const}}).$$

For a cyclic private output, it is

$$\mathrm{ReLU}\left(\frac{H+1}{2r} u_s^\top g_{\mathrm{phase}} + g_{\mathrm{token},z} - H g_{\mathrm{const}}\right).$$

Both reduce to the integer-margin selectors in the Euclidean construction. Weights $W_2 = x_{\mathrm{target}} - g$ use the actual variance-one vectors. QKV projections use token amplitude one; constants, however small, can use the permanent $+1$. All parameters are rational after fixing integer r.

C Payload preservation, delay and algebraic closure

C.1 Payload-preserving components

A component accepts a joint input label (z, p), where p is any finite payload. Lift its QKV projections to depend only on the old label z. The recurrent trajectory is unchanged. In the final conjunction selectors, index by the entire actual input label (z, p) and the exposed finite state. Choose W_2 to return the desired output together with (z, p) in one canonical label. This works for every primitive above. It also lifts each layer of a composed block: widths are chosen for the finite enlarged alphabets before choosing the datatype. No factorized-amplitude payload interface is assumed.

C.2 Flip-flop and one-step delay

For U_2 use its multiplication with a, b as right-zero resets. Keys select an inert row on identity and the active row on either reset; all gates remain one. The active row is zero before any reset, then holds one of e_a, e_b. A separate overwrite companion head, $k = q = e_0, v = 1$, supplies the current constant. The output projection forms $(1 - S_a - S_b, S_a, S_b)$, including the identity state, and the finite recoder returns the inclusive product while retaining the full input payload. Companion heads are counted in the common finite head count, rather than assumed to be available from the reset head.

After any lower block, run a cyclic-two block on constant increment one, preserving the current lower product and input, to expose $p_t = t \bmod 2$. A fresh buffer head uses $k_t = e_{p_t}, q_t = e_{1-p_t}, v_t = e_{N_t}$. Its two rows are initially zero. At time t the selected row is overwritten and the opposite row, most recently written at $t-1$, is read. Hence the readout is zero for $t = 1$ and $e_{N_{t-1}}$ afterwards, by induction. A separate current-value overwrite companion $k = q = e_0, v = (1, B_t)$ supplies the constant and first-position correction. Form $e_{N_{t-1}}$ by adding $B_t e_{1_N}$. The same paired carrier and recoder preserve all current and delayed labels. These companion channels do not rely on the buffer retaining current values.

C.3 Wreath products and divisors

Suppose blocks for M and N compute their full-alphabet prefix products and preserve every finite payload. Compute N_t, then N_{t-1} as above, retaining the current (f_t, n_t). For each fixed $x \in N$, a finite controller computes $f_t(xN_{t-1})$ as an input label to a fresh M block. Run these $|N|$ blocks sequentially in depth, preserving the lower states and previously computed upper products. Their fresh zero memories do not interfere. By induction on positions, the xth result is

$$F_t(x) = \prod_{i=1}^{t} f_i(xN_{i-1}),$$

which, with N_t, is precisely the prefix product in $M \wr N$. All additional components have fixed finite depth; no optimal sum-depth bound is required for the theorem.

If A is a homomorphic image of a submonoid $T \leq W$, choose a set-theoretic section $\sigma : A \to T$ of the surjective homomorphism $\pi : T \to A$. Run the full-alphabet W block on $\sigma(a_t)$. Every reached product lies in T, and

$$\pi\left(\prod_{i=1}^{t} \sigma(a_i)\right) = \prod_{i=1}^{t} a_i.$$

Only π is multiplicative; the section need not be. Implement the section and quotient decoding with the finite recoders above, extending the decoder arbitrarily outside T. The soluble Krohn–Rhodes decomposition into cyclic prime groups and U_2 factors, as stated in StateBench Proposition 1 (see also the classical decomposition in [5, Appendix I, Theorem I.3]), now completes a structural induction. The trivial factor is a constant recoder.

D A common datatype and all-length bounds

First fix the finite algebraic program, every joint alphabet and coordinate order, and a sufficiently large common width and head count. Unit carriers pad with zero coordinates. Variance carriers use Appendix B. Choose the common width divisible by the head count and large enough that every head block fits its active rows and value columns (for variance width $4r^2$, choose r a sufficiently large multiple of that count). Pad unused heads with unit key and zero query/value, so they have zero output and memory; do not normalize a dummy head in isolation. Fix feedforward width large enough for all finite conjunction selectors.

All carriers, finite-controller targets, selector margins and normalization intermediates now have fixed rational values, independent of D. Choose D to clear every denominator of these values and their individual ordinary products, including coordinates divided by n, coordinate squares and squares divided by n. Include factors 36 and m^2 for each cyclic modulus, and require $L_m = D/m^2$ divisible by the fixed denominator factors of its output coefficients. The finite list has a common multiple. Only after fixing this D do we set the L_m and instantiate the grid-dependent parameters. The theorem quantifies existentially over D, p and those parameters; it does not hold them fixed while arbitrarily changing the grid. The following audit separates intentionally rounded operations from ordinary exact ones.

1. Canonical projections, finite selectors, FF writes, binary reset/buffer states and the norm calculations have fixed rational operands/products. Clearing their individual denominators makes them exact, including both mean and variance evaluation orders checked by the certificates.

2. Cyclic keys and materialized transition entries are exact once $m^2 \mid D$. State numerators are mr; retained products have the intentionally rounded numerators derived in Lemma 1, independent of D. Their tiny writes have integer numerator ma. Overwrite companions cancel exact integer retained terms.

3. Marker transition entries have denominator 36, and state numerators belong to $\{0, u, v, w\}$. Its products with these $1/D$ states deliberately round, with the same numerator map at every allowed D. Values $1/D, 2/D$ are generated exactly from the canonical constants; clearing denominators is not asserted to remove the marker's intentional rounding.

4. Output coefficients proportional to D or L_m multiply state values proportional to $1/D$, producing the already fixed rational carriers. Their parameter grid numerators and individual product denominators impose only finitely many additional divisibility conditions. There is no requirement that a denominator depending recursively on D divide D. For example, a coefficient γD multiplying a state z/D yields the fixed rational γz. Its parameter numerator is γD^2; clearing the fixed denominator of γ suffices. Tiny marker values are generated from the canonical $+1/2$ (Euclidean) or permanent $+1$ (variance) coordinate, as in Appendix A; their intentional rounding is retained rather than eliminated.

Coverage under composition. Payload lifting replicates canonical projection columns and enlarges finite selector indices; it introduces only the products of case 1. Finite recoding, including sections, quotient decoding and the trivial factor, uses the overwrite head and residual-corrected FF of Appendix A: case 1. A parity clock reuses cases 2 and 4 and its marker uses case 3. Reset and delay heads use binary keys, queries, values and memories; their companion channels, first-position correction and carrier projections use cases 1 and 4. Wreath composition sequentially connects canonical labels and makes finitely many copies of these blocks; it does not multiply their memories together. Common-width padding adds zeros and inert unit-key heads, covered by case 1; variance padding uses the fixed rational geometry of Appendix B. In every block, gate factors are one, centering and squared-statistic products are in case 1, the root argument is exactly one, and division is by one. Residual and dot-product sums are ordered finite additions; ReLU, coordinate selection and the final argmax introduce no new rounded product. This structural induction covers the enlarged finite alphabets before D is chosen.

Each primitive has a finite memory invariant and finite canonical input alphabet. Their sequential product therefore has a finite reachable set, independent of input length. At fixed D there are finitely many parameters and finitely many exact pre-round scalar results and ordered partial sums in one transition. Let B bound their absolute physical values. Choose finite p with $2^p > DB$. Every transition stays inside the datatype range for every input length. This is an existence bound, not a precision-optimality or polynomial-size claim. The bound is chosen after D and includes parameters and ordered partial sums, not just final state entries. It provides no perturbation margin or floating-point equivalence.